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2025 Previous Year Questions

(Set-3)

BIOLOGY

Section-A

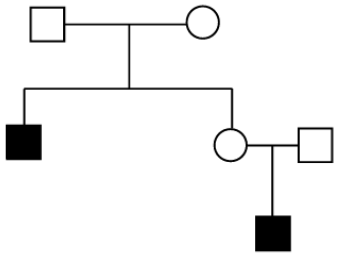
Questions no. 1 to 16 are Multiple Choice Type Questions, carrying 1 mark each. Choose the best option.

1. In its extended 'beads-on-string' form of chromatin, the 'beads' string represent:
(A) Linker DNA
(B) Histone proteins
(C) Nucleosomes
(D) NHC proteins

2. Given below are a few statements with reference to the accessory ducts of the human male reproductive system:
(i) The seminiferous tubules of the testes open into rete testis then into the vas deferens.
(ii) The vasa efferentia leave the testes and open into the epididymis.
(iii) The epididymis leads to vas deferens that ascends into the abdomen.
(iv) The vas deferens receives a duct from the prostrate gland and opens into the urethra as ejaculatory duct.
(v) The urethra originates from the urinary bladder and extends through the penis to its external opening, urethral meatus.
Choose the option with all true statements from the given options:
(A) (i), (ii), (iv)
(B) (ii), (iii), (v)
(C) (ii), (iv), (v)
(D) (i), (iii), (iv)

3. The substrate used during DNA replication by the enzyme DNA-dependent DNA polymerase is:
(A) Deoxyribonucleotide triphosphate
(B) Deoxyribonucleoside triphosphate
(C) Ribonucleotide triphosphate
(D) Ribonucleoside triphosphate

4. In the given pedigree chart, a cross between a normal couple resulted in a son who was haemophilic and a normal daughter. In course of time, when the daughter was married to a normal man, to their surprise the grandson was also haemophilic.



Choose the option that indicates the correct inheritance of trait in the above pedigree chart:
(A) Autosome linked dominant trait
(B) Sex-linked dominant trait
(C) Autosomal recessive trait
(D) Sex-linked recessive trait

5. In which of the following human diseases does the body's self-defence mechanism attack self-cells?
(A) Thalassemia
(B) Phenylketonuria
(C) Filariasis
(D) Rheumatoid arthritis

6. Select the statements that are true for a typical dicotyledonous embryo from the given options.
(i) It consists of an embryonal axis and scutellum.
(ii) The portion of embryonal axis above the level of cotyledon is epicotyl.
(iii) The portion of embryonal axis below the level of cotyledon is coleorhiza.
(iv) The lower end of the embryo has radicle covered with a root cap. Choose the correct answer:
(A) (i) and (ii)
(B) (i) and (iii)
(C) (iii) and (iv)
(D) (ii) and (iv)

7. About 15 mya during human evolution, the primates which used to walk like gorillas and chimpanzees were:

(A) *Australopithecine* and *Neanderthal*
(B) *Dryopithecus* and *Ramapithecus*
(C) *Homo erectus* and *Homo sapiens*
(D) *Homo habilis* and *Homo erectus*

8. Use the given information to select the amino acid attached to the 3' end of tRNA during the process of translation, if the coding strand of the structural gene being transcribed has the nucleotide sequence TAC.

Codons for the amino acids:
AUC Isoleucine
AUG Methionine
UAC Tyrosine
GUA Valine

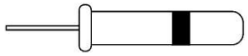
Options:

(A) Isoleucine (B) Methionine
(C) Tyrosine (D) Valine

9. The technique for the early detection of a disease based on the principle of antigen-antibody interaction is:

(A) RNAi (B) EST
(C) PCR (D) ELISA

10. The correct depiction of the experiment performed by Matthew Meselson and Franklin Stahl to prove that DNA replicates semi-conservatively on separation of DNA by centrifugation after 40 minutes is:

(A) 
(B) 
(C) 
(D) 

11. Bioactive molecule Cyclosporin A used for human welfare is derived from:

(A) Propionibacterium sharmanii
(B) Monascus purpureus
(C) Trichoderma polysporum
(D) Aspergillus niger

12. In a pea plant (Pisum sativum) inflated pod shape is dominant over constricted pod shape. The expected ratio of phenotypes of the offspring in a cross between both the parents with heterozygous inflated pod shape will be:

(A) 1:0 (B) 1:1
(C) 2:1 (D) 3:1

For Questions number 13 to 16, two statements are given one labelled as Assertion and the other labelled as Reason. Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

13. **Assertion (A):** The meristems are grown 'in vitro' to obtain virus-free plants from an infected plant.
Reason (R): If the plant is infected with a virus, the roots and the stems are free of virus.

14. **Assertion (A):** A person infected with malaria suffers from chill and high fever, recurring every three or four days.
Reason (R): The parasite attacks the RBC resulting in their rupture and release of haemozoin.

15. **Assertion (A):** 'Saheli', an oral contraceptive inhibits ovulation and increases phagocytosis of sperms.
Reason (R): It is a non-steroidal preparation.

16. **Assertion (A):** ABO blood grouping in humans is an example of multiple allelism.
Reason (R): More than two genes in a population govern the same character in ABO blood grouping in humans.

Section-B

17. (a) Explain how the immunity of a person is affected if there is atrophy (degeneration) of the thymus gland at an early stage of life.

OR

(b) (i) What are interferons? Explain their role in providing immunity to a person.
(ii) Which category of innate immunity defence barrier can interferons be classified into?

18. Assume that the given mRNA (start site is not depicted) is theoretically translated in two reading frames.

(a) Translation starting from the first nucleotide (Reading frame 1)
(b) Translation starting from the second nucleotide (Reading frame 2)

19. (a) Explain what is meant by the term MTP. What was the main reason to legalize MTP by the Government of India?

OR

(b) Name any two STIs which might occur in a human female. State its two early symptoms.

20. The basic scheme of the essential steps involved in the process of recombinant DNA technology is summarized below in the form of a flow diagram. Study the given flow diagram and answer the questions that follow.

Step-1 Vector DNA (cut using Restriction Enzyme EcoR I) + Alien DNA (cut using Restriction Enzyme EcoR I)

↓ Enzyme A

Step-2 Recombinant DNA molecule

↓

Step-3 Transfer of recombinant DNA molecule in *E. coli* (Host)

↓

Step-4 Replication of recombinant DNA molecule in *E. coli*

(a) Name the enzyme used in Step-1 to join the cut plasmid and alien DNA.
(b) State the technical term used for Step-3.
(c) Justify the use of same Restriction Enzyme EcoR I to cut both the vector DNA and the alien DNA.

21. (a) Explain how the interaction between sea anemone and clownfish is one of the best examples of commensalism in nature.

OR

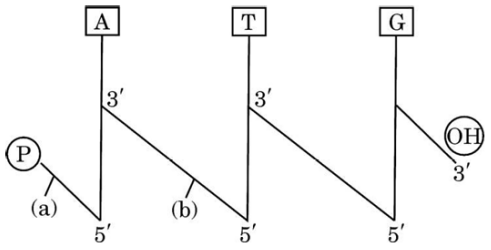
(b) Correctly depict (also indicate the trophic level) and describe the ecological pyramid of biomass in sea with 40 standing crop of phytoplankton supporting 90 standing crop of zooplankton which further supports 120 small fishes.

Section-C

22. Explain the process of formation of placenta in a human female after the implantation of the blastocyst in the endometrium of the uterus.

23. Gregor Mendel conducted hybridisation experiments in garden pea for seven years and proposed the law of inheritance.
(a) Why was he successful in his hybridisation experiments? Give two reasons.
(b) State the law of independent assortment as proposed by Mendel after his dihybrid crosses.

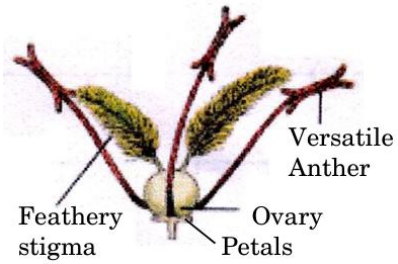
24. Study the given below single strand of deoxyribonucleic acid depicted in the form of a "stick" diagram with 5' – 3' end directionality, sugars as vertical lines and bases as single letter abbreviations and answer the questions that follow.



(a) Name the covalent bonds depicted as (a) and (b) in the form of slanting lines in the diagram.
(b) How many purines are present in the given "stick" diagram?
(c) Draw the chemical structure of the given polynucleotide chain of DNA.

25. Explain the biological treatment of primary effluent when passed into the large aeration tanks in a sewage treatment plant (STP).

26. Given below is a flower with its characteristic features specialised for the most common type of abiotic pollination.



Answer the following questions based on the above diagram:
(a) Name the mode of abiotic pollination that will be adopted by the given plant species in the above picture.
(b) State the need of exposed large feathery stigmas for the flower.
(c) What will be the two important adaptations in the pollen grains of the flowers pollinated by the above mode of pollination?
(d) What could be the probable reason for the petals being small and non-green?

27. According to a recent wildlife report, the biggest threat to the tiger's survival in Mudumalai Tiger Reserve (MTR) was found to be a small, beautiful flower, Lantana camara, a tropical American shrub, that invaded 40% of India's tiger range. Tamil Nadu department's Lantana weed eradication drive helped to restore the dying MTR thereby also reducing human-wildlife conflicts. MTR is home to 25 species of grasses and legumes.
Answer the given questions based on the information given above.
(a) Explain how did the removal of Lantana help in restoring the dying Mudumalai Tiger Reserve.
(b) Why is the invasion of Lantana camara a cause of concern in MTR.

28. Enlist one advantage and two disadvantages of green revolution.

Section-D

Questions No. 29 and 30 are case-based questions. Each question has 3 sub-questions with internal choice in one sub-question.

29. Read the following passage and answer the questions that follow.

The most convincing evidence to trace evolutionary relationships between humans and different groups of animals come from the basic similarities seen at the molecular level. Study the table given below that depicts the number of amino acid differences between the haemoglobin polypeptide of few animals with that of humans and answer the questions that follow.

S. No.	Animal	No. of amino acid differences in haemoglobin with that of humans
1.	Macaque	08
2.	Dog	32
3.	Bird	45
4.	Frog	67
5.	Lamprey	125

(a) To which category of evolution (Divergent or Convergent) do the following evolutionary relationships belong to:
  (i) Humans and Macaque
  (ii) Humans and Frog

(b) What do the biochemical similarities in haemoglobin suggest about the evolutionary relationship between humans, frog and lamprey?

(c) (i) Which one of the two – lampreys' or macaques' evolution is more closely related to humans and why?

OR

(ii) Which one of the two – frogs' or dogs' evolution is more closely related to humans and why?

30. Read the following passage and answer the questions that follow.

Deaths related to the use of drugs were estimated at about 5,00,000 in 2019, 17-5 percent more than in 2009. Liver diseases attributed to Hepatitis B are a major cause of drug-related deaths, according to UNODC, accounting for more than half of the total number of deaths attributed to the use of drugs. Drug overdoses account for a quarter of drug-related deaths.

Opioids contribute to account for the most severe drug-related harm, including fatal overdoses, when used non-medically. At the global level, two-third of direct drug-related deaths are due to opioids, and in some sub-regions the proportion can be as high as three-quarters of such deaths.

(a) Why are people taking opioids more prone to liver diseases attributed to Hepatitis B?

(b) What is meant by direct drug-related disease?

(c) (i) What is the scientific name of the plant from which the opioids are derived and from which part of the plant is it extracted?

OR

(c) (ii) State two common warning signs of drug abuse among the youth.

Section-E

31. (a) (i) Explain how does double fertilisation take place in a flowering plant.

(ii) Write the fate of the products of double fertilization in these plants.

OR

(i) Explain the structure of testicular lobules in human male reproductive system. Name the two types of cells present in the seminiferous tubules and state their role.

(ii) Describe the role of hypothalamic hormone GnRH in spermatogenesis.

32. (a) Name and explain the biotechnological strategy wherein the infection by the nematode Meloidegyne incognitia can be prevented using Agrobacterium vectors in the roots of tobacco plant by RNA interference.

OR

(b) Explain the amplification of gene of interest using the technique of Polymerase chain reaction (PCR).

33. (a) (i) Describe the population growth curve applicable in a population of any species in nature that has limited resources at its disposal.

(ii) Give the equation of this growth curve.

(iii) Name the growth curve and depict a graphical plot for this type of population growth.

OR

(b) (i) Explain the Species-Area relationship within a natural forest and also predict the nature of graph when species richness is plotted against the area for a wide variety of taxa.

(ii) Depict the graphical relationship between species richness and area.

(iii) Give the equation of the Species-Area relationship for a wide variety of taxa on a logarithmic scale.

■■■

SPACE FOR ROUGH WORK

[5]

2024 Previous Year Questions

(Set-2)

BIOLOGY

Section-A

Question Nos. 1 to 16 are multiple choice type questions, carrying 1 mark each.

1. A single gene that controls the expression of more than one trait is said to show.

(A) Multiple allelism
(B) Polygenic inheritance
(C) Incomplete dominance
(D) Pleiotropism

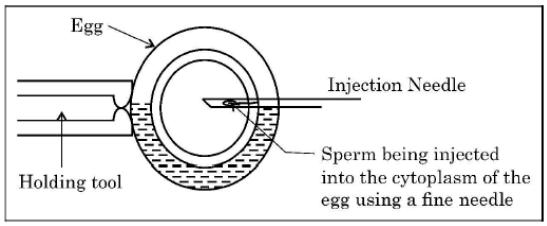
2. A person with trisomy of 21st chromosome shows.

(i) Furrowed tongue
(ii) Characteristic palm crease
(iii) Rudimentary ovaries
(iv) Gynaecomastia

Select the correct option, from the choices given below:

(A) (ii) and (iv) (B) (i), (ii) and (iv)
(C) (ii) and (iii) (D) (i) and (ii)

3. Observe the schematic representation of assisted reproductive technology given below:



Identify the most appropriate technique depicted in the above diagram.

(A) IUT (B) IUI
(C) ICSI (D) ZIFT

4. Interferons are proteins secreted by

(A) RBC
(B) WBC
(C) Bacteria infected cell
(D) Virus infected cell

5. During biological treatment of sewage, the masses of bacteria held together by fungal filaments to form mesh like structures are called

(A) primary sludge (B) flocs
(C) activated sludge (D) anaerobic sludge

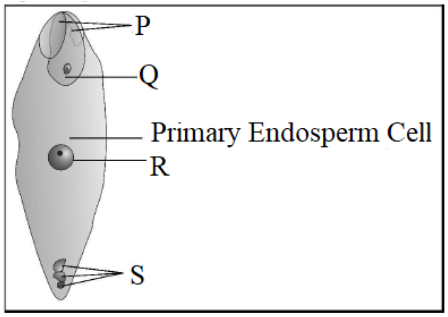
6. Which one of the following statements is correct in the context of observing DNA separation by agarose gel electrophoresis?

(A) DNA can be seen in visible light.
(B) DNA can be seen without staining in visible light.
(C) Ethidium bromide stained DNA can be seen in visible light.
(D) Ethidium bromide stained DNA can be seen under UV light.

7. A phenomenon where a male insect mistakenly identified the patterns of a orchid flower as the female insect partner, and tries to copulate and thereby pollinates the flower is said to be:

(A) Pseudocopulation
(B) Pseudopollination
(C) Pseudoparthenocarpy
(D) Pseudofertilisation

8. Identify the correct labellings in the figure of a fertilised embryo sac of an angiosperm given below:



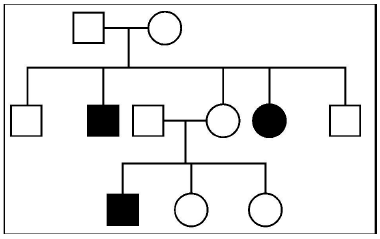
(A) P – zygote, Q – degenerating synergids, R – degenerating antipodals, S – PEN

(B) P – degenerating synergids, Q – zygote, R – PEN, S – degenerating antipodals

(C) P – degenerating antipodals, Q – PEN, R – degenerating synergids, S – zygote

(D) P – degenerating synergids, Q – zygote, R – degenerating antipodals, S – PEN

9. Study the pedigree chart of a family showing the inheritance pattern of a certain disorder. Select the option that correctly identifies the nature of the trait depicted in the pedigree chart.



(A) Dominant X – linked

(B) Recessive X – linked

(C) Autosomal dominant

(D) Autosomal recessive

10. Match the following genes of the lac operon listed in column 'I' with their respective products listed in column 'II':

	Column I (Gene)		**Column II (Products)**
P.	'i' gene	(i)	β-galactosidase
Q.	'z' gene	(ii)	lac permease
R.	'a' gene	(iii)	repressor
S.	'y' gene	(iv)	transacetylase

Select the correct option:

(A) P-(i), Q-(iii), R-(ii), S-(iv)

(B) P-(iii), Q-(i), R-(ii), S-(iv)

(C) P-(iii), Q-(i), R-(iv), S-(ii)

(D) P-(iii), Q-(iv), R-(i), S-(ii)

11. If both the parents are carriers for thalassaemia, the chances of an afflicted child to be born to them is:

(A) 25% (B) 50%

(C) 75% (D) 100%

12. If the sequence of nitrogen bases of the coding strand in a transcription unit is 5' – ATGAATG – 3', the sequence of bases in its RNA transcript would be

(A) 5' – AUGAAUG – 3'

(B) 5' – UACUUAC – 3'

(C) 5' – CAUUCAU – 3'

(D) 5' – GUAAGUA – 3'

For Questions number 13 to 16, two statements are given one labelled as Assertion and the other labelled as Reason. Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion.

(B) Both Assertion and Reason are true, but Reason is not the correct explanation of the Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

13. **Assertion:** AIDS is a syndrome caused by HIV.
Reason: HIV is a virus that damages the immune system with DNA as its genetic material.

14. **Assertion:** In molecular diagnosis, single stranded DNA or RNA tagged with radioactive molecule is called a probe.
Reason: A probe always searches and hybridises with its complementary DNA in a clone of cells.

15. **Assertion:** In birds the sex of the offspring is determined by males.
Reason: Males are homogametic while females are heterogametic.

16. **Assertion:** Communities that comprise of more species tend to be more stable.
Reason: A higher number of species results in less year to year variation in total biomass.

Section-B

17. (a) "Farmers prefer apomictic seeds to hybrid seeds." Justify giving two reasons.

OR

(b) Mention one advantage and one disadvantage of amniocentesis.

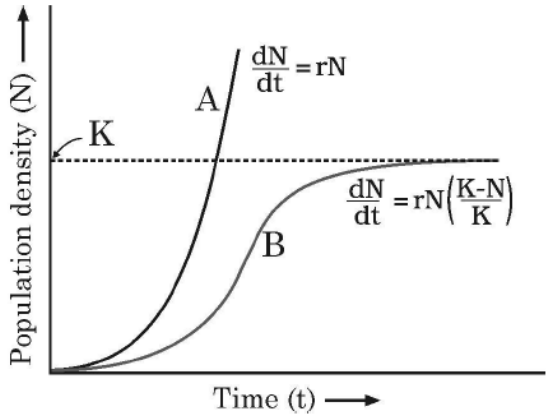
18. 5' – G↓AATTC – 3'

3' – CTTAA↑, G – 5'

(a) Name the restriction enzyme that recognises the given specific sequence of bases. What are such sequence of bases referred to as?

(b) What are the arrows in the given figure indicating? Write the result obtained thereafter.

19. Observe the population growth curve and answer the questions given below:



(a) State the conditions under which growth curve 'A' and growth curve 'B' plotted in the graph are possible.

(b) Mention what does 'K' in the graph represent.

20. Explain how are plants benefitted by their association with "*Glomus* species".

21. If the base adenine constitutes 31% of an isolated DNA fragment, then write what will be the expected percentage of the base cytosine in it. Explain how did you arrive at the answer given.

Section-C

22. Identify a, b, c, d, e and f in the table given below:

Sl. No.	Organism	Bioactive Molecule	Use
1.	*Monascus purpureus*	a	b
2.	c	d	Antibiotic
3.	e	Cyclosporin A	f

23. (a) Tropical regions harbour more species than the temperate regions. How have biologists tried to explain this in their own ways? Explain.

OR

(b) (i) What does an ecological pyramid represent?

(ii) The Ecological pyramids may have an 'upright' or an 'inverted' shape. Justify with the help of suitable examples.

24. (a) What are transgenic animals?

(b) Name the transgenic animal having the largest number amongst all the existing transgenic animals.

(c) State any 3 reasons for which these types of animals are being produced.

25. If the cells in the leaves of a maize plant contain 10 chromosomes each, write the number of chromosomes in its endosperm and zygote. Name and explain the process by which an endosperm and a zygote are formed in maize.

26. (a) Why does DNA replication occur within a replication fork and not in its entire length simultaneously?

(b) "DNA replication is continuous and discontinuous on the two strands within the replication fork." Explain with the help of a schematic representation.

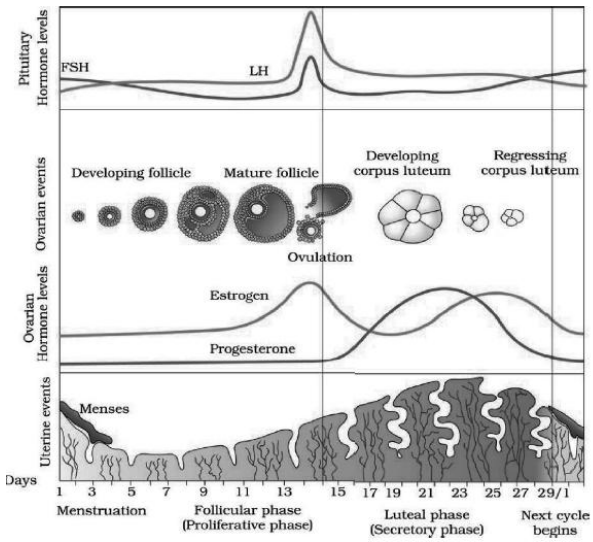
27. Explain the processing of heterogeneous nuclear RNA (hnRNA) into a fully functional mRNA in eukaryotes. Where does this processing occur in the cell?

28. The world is facing accelerated rates of species extinction largely due to human activities. Explain any three human activities responsible for accelerated rates of species extinction.

Section-D

Q. No. 29 and 30 are case based questions. Each question has 3 sub-questions with internal choice in one sub-question.

29. In a human female, the reproductive phase starts on the onset of puberty and ceases around middle age of the female. Study the graph given below regarding menstrual cycle and answer the questions that follow:



(a) Name the hormones and their source organ, which are responsible for menstrual cycle at puberty.

(b) For successful pregnancy, at what phase of the menstrual cycle an early embryo (upto 3 blastomeres) should be Implanted in the Uterus (IUT) of a human female who has opted for Assisted Reproductive Technology (ART)? Support your answer with a reason.

(c) Name the hormone and its source organ responsible for the events occurring during proliferative phase of menstrual cycle. Explain the event.

OR

In a normal human female, why does menstruation only occurs if the released ovum is not fertilised? Explain.

30. Read the following passage and answer the questions that follow:

"Mosquitoes are drastically affecting the human health in almost all the developing tropical countries. Different species of mosquitoes cause very fatal diseases so much so that many humans loose their life and if they survive, are unable to put in productive hours to sustain their life. With the result the health index of the country goes down."

(a) Name the form in which *Plasmodium* gains entry into (i) human body (ii) the female Anopheles body.

(b) Why do the symptoms of malaria not appear in a person immediately after being bitten by an infected female *Anopheles*? Give one reason. Explain when and how do the symptoms of the disease would appear.

OR

Explain the events which occur within a female *Anopheles* mosquito after it has sucked blood from a malaria patient.

(c) Name a species of mosquito other than female *Anopheles* and the disease, for which it carries the pathogen.

Section-E

31. (a) (i) Draw a schematic diagram of the cloning vector PBR 322 and label (1) Bam HI site (2) gene for ampicillin resistance (3) 'ori' (4) 'rop' gene.

(ii) State the role of 'rop' gene.

(iii) A cloning vector does not have a selectable marker. How will it affect the process of cloning?

(iv) Why is insertional inactivation preferred over the use of selectable markers in cloning vectors?

OR

(B) (i) Name the nematode (scientific name) that infects the roots of tobacco plant and reduces its yield.

(ii) Name the vector that is used to introduce nematode-specific genes into the host plant (tobacco).

(iii) How do sense and anti-sense RNAs function?

(iv) Why could parasite not survive in a transgenic tobacco plant?

32. (a) (i) Draw a diagram of a human sperm. Label any four parts and write their functions.

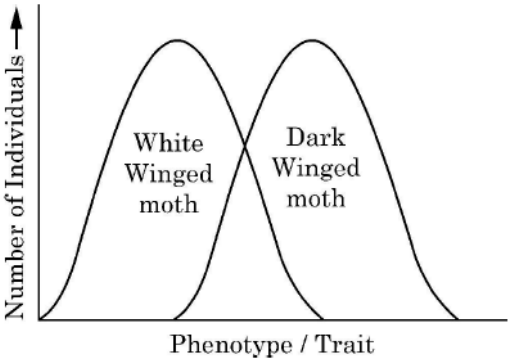
(ii) In a human female, probability of an ovum to get fertilized by more than one sperm is impossible. Give reason.

OR

(b) (i) With the help of labelled diagram only, show the different stages of embryo development in a dicot plant.

(ii) Endosperm development precedes embryo development. Justify.

33.



(a) Natural selection operates in different ways in nature.

(i) Identify the type of natural selection depicted in the graph above.

(ii) In England after industrialisation, the population of dark winged moths were more favoured than white winged moth. Explain.

(iii) Anthropogenic action can enhance the rate of evolution. Explain with the help of an example.

OR

(b) (i) Why did Hershey and Chase use '^{35}S' and '^{32}P' in their experiment? Explain.

(ii) State the importance of (1) blending and (2) centrifugation in their experiment.

(iii) Write the conclusion they arrived at the end of their experiment.

■■■

SPACE FOR ROUGH WORK

[6]

2023 Previous Year Questions

(Set-3)

BIOLOGY

Section-A

1. At which stage during evolution did human use hides to protect their bodies and buried their dead?
(a) Homo habilis (c) Java man
(b) Neanderthal man (d) Homo erectus

2. Given below are Column A with a list of certain Assisted Reproductive Technologies (ART) and in Column B the procedures followed during ART:

Column A		Column B	
S. No.	Names of ART	S. No.	Procedures
(A)	GIFT	(i)	Transfer of ovum from a donor into the fallopian tube of another female.
(B)	ICSI	(ii)	Transfer of semen from the donor into the vagina of the female.
(C)	ZIFT	(iii)	Injecting sperms directly into the ovum.
(D)	IUI	(iv)	Transfer of early embryos into the fallopian tube.

Choose the option where ART correctly matches with the procedure.
(a) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
(b) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)
(c) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)
(d) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

3. The decrease in the T-lymphocytes count in human blood will result in:
(a) Decrease in antigens
(c) Increase in antibodies
(b) Decrease in antibodies
(d) Increase in antigens

4. Given below is a sequence of bases in mRNA of a bacterial cell. Identify the amino acid that would be incorporated at codon position 3 and codon position 5 during the process of its translation.
3' AUCAGGUUUGUGAUGGUACGA 5'
(a) Phenylalanine, Methionine
(b) Cysteine, Glycine
(c) Alanine, Proline
(d) Serine, Valine

5. A Tight one-to-one relationship between many species of fig tree and certain wasps is an example of-
(a) Commensalism (b) Parasitism
(c) Amensalism (d) Mutualism

6. Select the pathogen mismatched with the symptoms of disease caused by it from the list given below:
(a) *Entamoeba histolytica:* Constipation, abdominal pain.
(b) *Epidermophyton*: Dry scaly lesions on nail.
(c) *Wuchereria bancrofti*: Chronic inflammation of lymphatic vessels of lower limb.
(d) *Haemophilus influenzae*: Blockage of the intestinal passage.

7. The primary productivity in an ecosystem is expressed as:
(a) $gm^{-2}\ yr^{-1}$ (b) $gm^{-2}\ yr$
(c) K cal $m^{-2}\ yr^{-1}$ (d) K cal m^{-2}

8. Given below is the restriction site of a restriction endonuclease Pst-I and the cleavage sites on a DNA molecule.

5' C – T – G – C – A ↓ – G 3'
3' G ↑ – A – C – G – T – C 5'

Choose the option that gives the correct resultant fragments by the action of the enzyme Pst-I.
(a) 5' C – T – G C – A – G 3'
3' G – A – C – G – T C5'
(b) 5' C – T G – C – A – G 3'
3' G – A – G – C T – C 5'
(c) 5' C – T – G – C A – G 3'
3' G – A – C – G T – C 5'
(d) 5' C – T – G – C – A C – A – G 3'
3' G A – C – G – T – C 5'

9. The IUCN Red Data List (2004) in the last 500 years documents the extinction of nearly 784 species including:
(a) 330 invertebrates
(b) 338 invertebrates
(c) 359 invertebrates
(d) 362 invertebrates

10. Given below are the list of the commercially important products and their source organisms. Select the option that gives the correct matches.

List A		List B	
S. No.	**Bioactive Products**	**S. No.**	**Microbes (Source Organism)**
(A)	Cyclosporin A	(i)	*Streptococcus*
(B)	Statins	(ii)	*Tricoderma polysporum*
(C)	Streptokinase	(iii)	*Penicillium notatum*
(D)	Penicillin	(iv)	*Monascus purpureus*

Options:

(a) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
(b) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)
(c) (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)
(d) (A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)

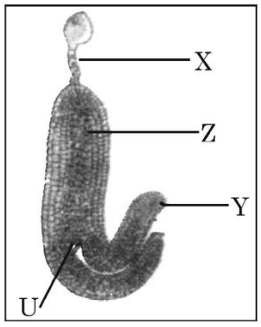
11. Important attributes belonging to a population but not to an individual are:

(i) Birth rate and death rate
(ii) Male and female
(iii) Birth and death
(iv) Sex-ratio

Select the correct option from the given options:

(a) (i) only (b) (ii) only
(c) (ii) and (iii) (d) (i) and (iv)

12. Select the option that shows the correctly identified 'U', 'X', 'Y' and 'Z' in a developing dicot embryo.



(a) X – Plumule (2n), Y – Suspensor (n), Z – Cotyledon (2n), U – Radicle (2n).
(b) X – Plumule (2n), Y – Suspensor (2n), Z – Radicle (2n), U – Cotyledon (2n).
(c) X – Suspensor (2n), Y – Cotyledon (2n), Z – Radicle (2n), U – Plumule (2n).
(d) X – Cotyledon (2n), Y – Radicle (n), Z – Plumule (n), U — Suspensor (n).

For Questions number 13 to 16, two statements are given one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion and Reason are true and Reason is the correct explanation of the Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation of the Assertion.
(c) Assertion is true, but Reason is false.
(d) Assertion is false, but Reason is true.

13. **Assertion (A):** Determining the sex of an unborn child followed by MTP is illegal practice.
Reason (R): Amniocentesis is a practice to test the presence of genetic disorders also.

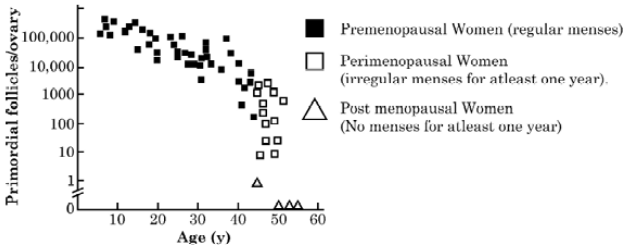
14. **Assertion (A):** Synthetic oligonucleotide polymers are used during Annealing in a PCR.
Reason (R): The primers bind to the double stranded DNA at their complementary regions.

15. **Assertion (A):** Decomposition process is slower if detritus is rich in lignin and cutin.
Reason (R): Decomposition is largely an oxygen requiring process.

16. **Assertion (A):** In Thalassemia an abnormal myoglobin chain is synthesized due to a gene defect.
Reason (R): α-Thalassemia is controlled by genes HBA1 and HBA2 on chromosome 16.

Section-B

17. The graph given below shows the number of primordial follicles per ovary in women at different ages. Study the graph and answer the questions that follow.



(a) What is the average age of the women at the onset of menopause?
(b) At What age are maximum primordial follicles present in the ovary, according to the given graph?

18. "Cattle and goats do not browse the Calotropis plant." Justify the statement giving reasons.

19. By using Punnett square depict the genotypes and phenotypes of test crosses (where green pod colour (G) is dominant over yellow pod colour (g)) in Garden pea with unknown genotype.

20. (a) (i) Give an example of a genus of virus used as narrow spectrum insecticidal biocontrol agent.
(ii) How does its use serve as an aid in overall integrated pest management programme?

OR

(b) Why a malignant tumour considered to be more damaging than a benign tumour? Explain.

21. (a) Write the scientific name of the source organism of the thermostable DNA polymerase used in PCR.

OR

(b) State the advantage of using Thermostable DNA polymerase.

Section-C

22. Name and explain a surgical contraceptive method that can be adopted by the male partner of a couple.

23. Human Genome Project (HGP) was a mega project launched in the year 1990 with some important goals.

(a) Enlist any four prime goals of HGP.

(b) Name any one common non-human animal model organism which has also been sequenced thereafter.

24. Industrial melanism in England after 1850 is an excellent example of Natural selection. Explain how?

25. One of the major approaches of crop improvement programme is Artificial Hybridisation. Explain the Steps involved in making sure that only the desired pollen grain pollinate the stigma of a bisexual flower by a plant breeder.

26. (a) "*Plasmodium* protozoan needs both a mosquito and a human host for its continuity." Explain.

(b) We all must work towards maintaining good health because 'health is wealth'. Enlist any six ways of achieving good health.

27. On spraying Bacillus thuringiensis on an infected cotton crop field the pests are killed by the toxin, however the toxin although produced by the bacteria does not affect it. Explain giving reason.

28. "Biodiversity plays a major role in many ecosystem services that nature provides."

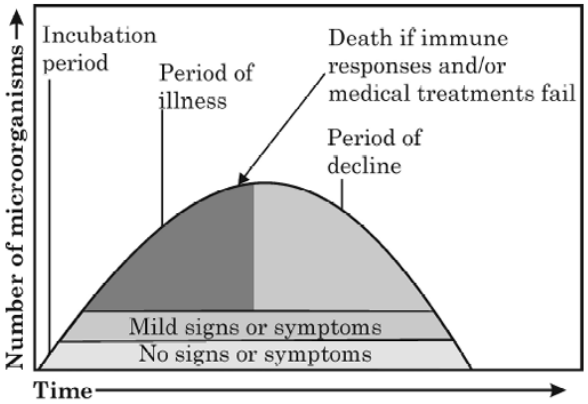
(a) Describe any two broadly utilatarian arguments to justify the given statement.

(b) State one ethical reason of conserving biodiversity.

Section-D

Q. Nos. 29 and 30 are case based questions. Each question has subparts with internal choice in one subpart.

29. When a microorganism invades a host, a definite sequence of events usually occur leading to infection and disease, causing suffering to the host. This process is called pathogenesis. Once a microorganism overcomes the defense system of the host, development of the disease follows a certain sequence of events as shown in the graph. Study the graph given below for the sequence of events leading to appearance of a disease and answer the questions that follow:



(a) In which period, according to the graph there are maximum chances of person transmitting a disease / infection and why?

(b) Study the graph and write what is an incubation period. Name a sexually transmitted disease that can be easily transmitted during this period. Name the specific type of lymphocytes that are attacked by the pathogen of this disease.

OR

(b) Draw a schematic labelled diagram of an antibody.

(c) In which period, the number of immune cells forming antibodies will be the highest in a person suffering from pneumonia?
Name the immune cells that produce antibodies.

30. The chromosome number is fixed for all normal organisms leading to species specification whereas any abnormality in the chromosome number of an organism results into abnormal individuals. For example, in humans 46 is the fixed number of chromosomes both in male and female. In male it is '44 + XY' and in female it is '44 + XX'. Thus the human male is heterogametic, in other words produces two different types of gametes one with '22 + X' chromosomes and the other with '22 + Y' chromosomes respectively. Human female, on the other hand is homogametic i.e. produces only one type of gamete with '22 + X' chromosomes only.

Sometimes an error may occur during meiosis of cell cycle, where the sister chromatids fail to segregate called nondisjunction, leading to the production of abnormal gametes with altered chromosome number. On fertilisation such gametes develop into abnormal individuals.

(a) State what is aneuploidy.

(b) If during spermatogenesis, the chromatids of sex chromosomes fail to segregate during meiosis, write only the different types of gametes with altered chromosome number that could possibly be produced.

(c) A normal human sperm (22 + Y) fertilises an ovum with karyotype '22 + XX'. Name the disorder the offspring thus produced would suffer from and write any two symptoms of the disorder.

OR

(c) Name a best known and most common autosomal aneuploid abnormality in human and write any two symptoms.

Section-E

31. (a) (i) Explain the monosporic development of embryo sac in the ovule of an angiosperm.

(ii) Draw a diagram of the mature embryo sac of an angiospermic ovule and label any four parts in it.

OR

(b) (i) Explain the formation of placenta after the implantation in a human female.

(ii) Draw a diagram showing human foetus within the uterus and label any four parts in it.

32. (a) Name and describe the steps involved in the technique widely used in forensics that serves as the basis of paternity testing in case of disputes.

OR

(b) It is sometimes observed that the F_1 progeny has a phenotype that does not resemble either of the two parents and has intermediate phenotype. Explain by taking a suitable example and working out the cross upto F_2 progeny.

33. (a) Bioreactors are the containment vehicles of any biotechnology-based production process. For large scale production and for economic reasons the final success of biotechnological process depends on the efficiency of the bioreactor.

Answer the following questions w.r.t. the given paragraph:

(i) List the operational guidelines that must be adhered to so as to achieve optimisation of the bioreactor system. Enlist any four.

(ii) Mention the phase of the growth we refer to in the statement "Optimisation of growth and metabolic activity of the cells".

(iii) Is the biological product formed in the bioreactor suitable for the intended use immediate? Give reason in support of your answer.

OR

(b) (i) 'EcoRI' has played very significant role in r-DNA technology.

(I) Explain the convention for naming EcoRI.

(II) Write the recognition site and the cleavage sites of this restriction endonuclease.

(ii) What are the protruding and hanging stretches of DNA produced by these restriction enzymes called? Describe their role in formation of r-DNA.

■■■

SPACE FOR ROUGH WORK

[5]

2022 Previous Year Questions

(Set-3)

BIOLOGY

Section-A

1. A boy developed some allergic reactions when he straight entered into his air conditioned room after a game of football outside his house. Write any two symptoms that could be noticed in such condition. How docs our body combat such conditions?

2. (a) (i) Write the Scientific name of the plant from where natural cannabinoids are obtained.
(ii) Mention the parts of the plant that are used for extracting the drug.
(iii) How does the drug affect human body?

OR

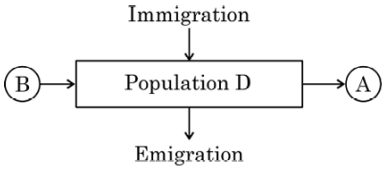
(b) Epithelial lining of our intestine is considered as secondary lymphoid organ. Justify the statement.

3. (a) Farmers are often suggested to use the following organisms in their crop land so as to improve the soil fertility.
(i) *Rhizobium* (ii) *Anabaena*
Explain.

4. Organic farmer use Trichoderma and Baculovirus as biological control agents. Explain.

5. Mammals are capable of maintaining homeostasis. But, very few smaller animals can survive in polar regions. Give reasons.

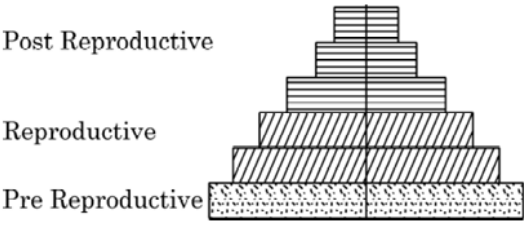
6. (a)



Observe the schematic representation given above and answer the following questions:
(i) Identify A and B.
(ii) Calculate the growth rate of bacteria in a curd sample, where 1 million bacteria increased to two million, within a period of one hour.

OR

(b)



Identify the type of pyramid given above. Write the identifying feature on the basis of which you identified it.

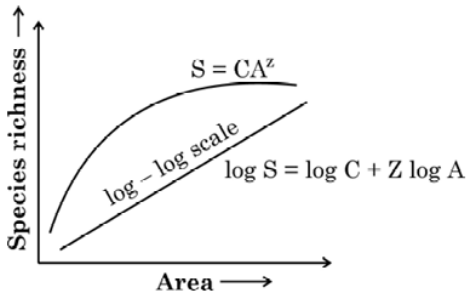
Section-B

7. (a) Write the complete name of the diagnostic test for AIDS. Explain the principle it works on.
(b) Name the type of genetic material present in AIDS causing pathogen.

8. A patient complains of suffering from constipation, stomach ache, stool with blood clots and excess mucous. The physician diagnosed it as amoebiasis, after stool test.
(a) Write the scientific name of the microbe identified in the stool sample.
(b) How do you think, the patient must have contracted it?
(c) Write your suggestions to the patient to avoid infection in future.

9. What are cry-proteins? With the help of a suitable example, explain how it acts as a biological pesticide.

10. (a) Write the inference drawn by Alexander von Humboldt after his extensive explorations of South Americal jungle.
(b) Study the graph given below:



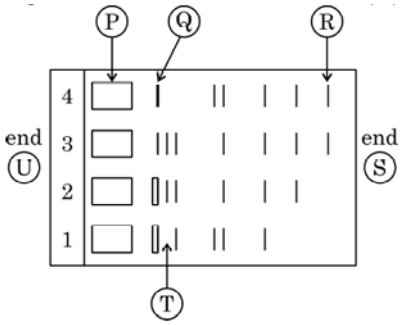
As per Alexander von Humboldt, what do the symbols S, A, Z and C in the graph stand for, in respect of a species and area relationship?

11. (a) Explain the concept of "co-extinction" by taking two examples.

OR

(b) "Forests provide intangible benefits to us." Explain by taking three different areas, how.

12. (a) Given below is the stepwise schematic representation of the process of electrophoresis. Identify the 'alphabets' representing (i) Anode end (ii) smallest/lightest DNA strand in the matrix (iii) Agarose gel



(b) What is elution? State the importance of elution in this process.

Section-C

13. (a) Read the paragraph given below and answer the questions that follow:

Enzyme Taq polymerase, is extracted from a eubacterial microorganism Thermus aauaticus from Yellowstone National Park in Montana, USA and isolated by Chien et al. (1976). Taq polymerase successfully replaced the DNA polymerase from E.coli that was being used in PCR earlier and this shift revolutionised the PCR technique.

(i) Taq polymerase after its discovery replaced E.coli DNA polymerase in PCR technique. Explain giving reasons why was the need felt for the change?

(ii) What is a primer and its importance in PCR?

(iii) Write the importance of PCR as a diagnostic tool.

OR

(b) Read the following paragraph and answer the questions that follow:

Biotechnology revolves around the "gene of interest", with an objective to open various avenues for human welfare in health, medicine, pharma, agriculture etc. using different techniques, tools and processes. One of the breakthroughs of biotechnology in medicine is the gene therapy.

(i) Name the human disease for which the gene therapy was used for the first time.

(ii) Explain the steps of gene therapy carried to cure the disease using the lymphocytes of the patient. Why is this therapy not a permanent cure of the disease?

(iii) Write the possible permanent cure of the disease by the gene therapy that is in progress.

■■■

SPACE FOR ROUGH WORK

[3]